Unit 2 Seminar Activity: Exploratory Data Analysis

Introduction

After the Ames tutorial, the five tasks ran on the 398-row Auto MPG file (Quinlan, 1993).

Findings by task

Missing values. A default pandas load reported no gaps: six horsepower entries hold '?', so it loaded as text. Parsed (Table 1, Figure 1), they were left missing; a median fill moved its mpg correlation by 0.005, bias unresolved. Ames has NA in 19 columns, but PoolQC's 99.5 per cent codes no pool, not a gap (De Cock, 2011).

Skewness and kurtosis. All seven skewness estimates are positive; model year is near zero (0.01); horsepower has the highest skewness (1.09) and excess kurtosis (0.70). SalePrice skews 1.88; Géron's (2025, ch. 2) log remedy cut that to 0.12; accuracy untested.

Correlation and scatter plots. Weight, displacement and horsepower correlate with mpg at -0.78 or stronger and mutually at 0.86 or more (Figure 2); collinearity to assess before regression (Géron, 2025, ch. 2). The horsepower scatter (Figure 3) suggests curvature: Spearman's rho -0.85 against Pearson's -0.78 (n = 392).

Encoding. Origin arrives coded 1, 2, 3; labels were restored and one-hot encoded, since linear and distance-based models read integers as ordered (Géron, 2025, ch.

2). Median mpg (Figure 4): Japan 31.6, America 18.5, a gap a linear weight adjustment narrows to 1.2 (Table 1).

Task	Result	Note
Ames tutorial	train.csv 1,460 × 81; test.csv 1,459 × 80 without SalePrice, so local evaluation needs a validation split from train.csv	The combined notebook's 47 code cells ran clean after deprecated or removed calls (np.object, distplot, factorplot) were replaced
Missing values	6 of 398 horsepower entries (1.5 per cent), stored as '?'	Left missing; horsepower correlations use the 392 available pairs
Median-fill sensitivity	horsepower to mpg r -0.778 (n = 392) against -0.773 (n = 398, median 93.5)	Difference 0.005; possible missingness bias remains unresolved
Skewness, excess kurtosis	mpg 0.46, -0.51; cylinders 0.53, -1.38; displacement 0.72, -0.75; horsepower 1.09, 0.70; weight 0.53, -0.79; acceleration 0.28, 0.42; model year 0.01, -1.18	Sample skewness; Fisher kurtosis (normal = 0); n = 398, horsepower 392; model year: near-zero skew, negative excess kurtosis
Ames SalePrice	skew 1.88, excess kurtosis 6.54; log SalePrice skew 0.12	Training set, n = 1,460
Correlation with mpg	weight -0.83; displacement -0.80; horsepower -0.78; cylinders -0.78; model year +0.58; acceleration +0.42	Pearson r, n = 398 (horsepower 392); Spearman: weight -0.88, displacement -0.86, horsepower -0.85
Strongest input pairs	cylinders to displacement 0.95; displacement to weight 0.93; displacement to horsepower 0.90	Shared variance among the engine measures
Origin encoding	America 249 cars, median mpg 18.5, mean weight 3,362 lb; Europe 70, 26.5, 2,423 lb; Japan 79, 31.6, 2,221 lb	Codes 1, 2, 3 restored to labels and mapped back as the brief asks; one-hot columns origin_America, origin_Europe, origin_Japan, each row summing to 1; median residual mpg after a linear weight fit: America -0.3, Europe -2.2, Japan +0.9,

		a descriptive adjustment, not a causal explanation

Table 1: Auto MPG results; every figure comes from a clean kernel restart with all cells run. Units: mpg in miles per US gallon, weight in pounds, displacement in cubic inches. Sources: the 398-row Auto MPG file (Quinlan, 1993) and the 1,460-row training split of the Ames data (original data De Cock, 2011)

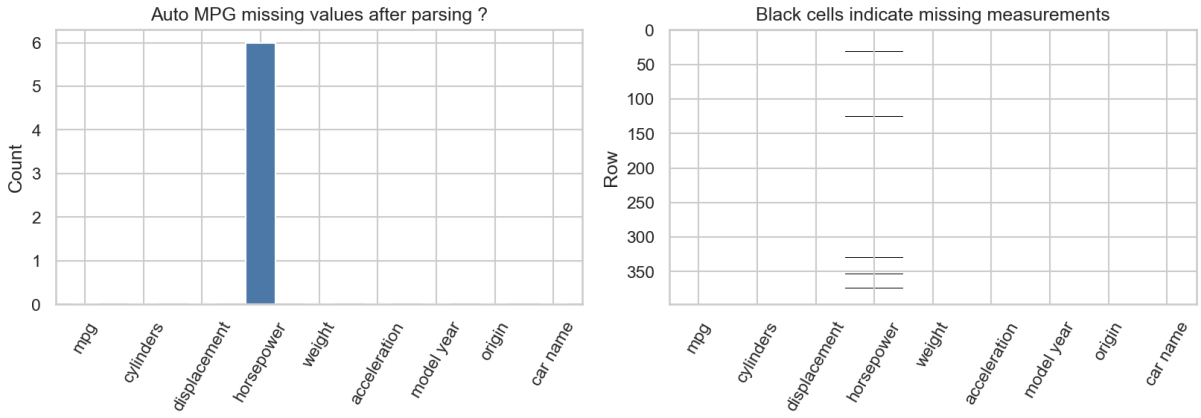


Figure 1: Missing values after parsing the '?' token; only horsepower has gaps, in six rows

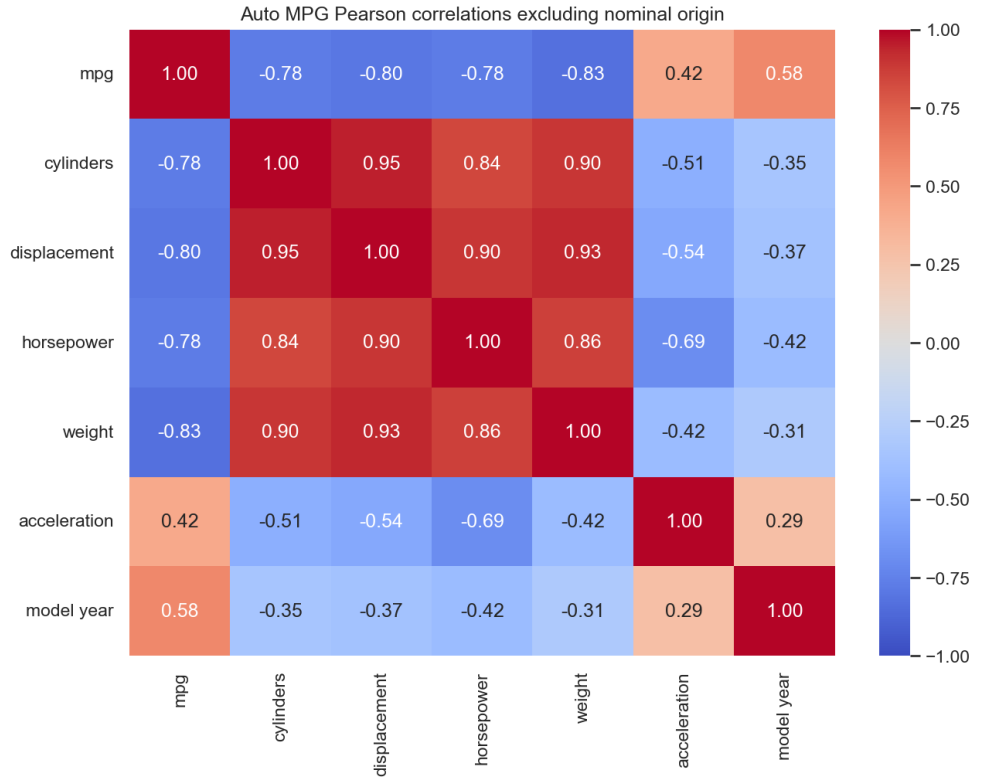


Figure 2: Pearson correlations among the numeric Auto MPG columns; origin excluded as nominal, cylinders and model year kept as ordered integers

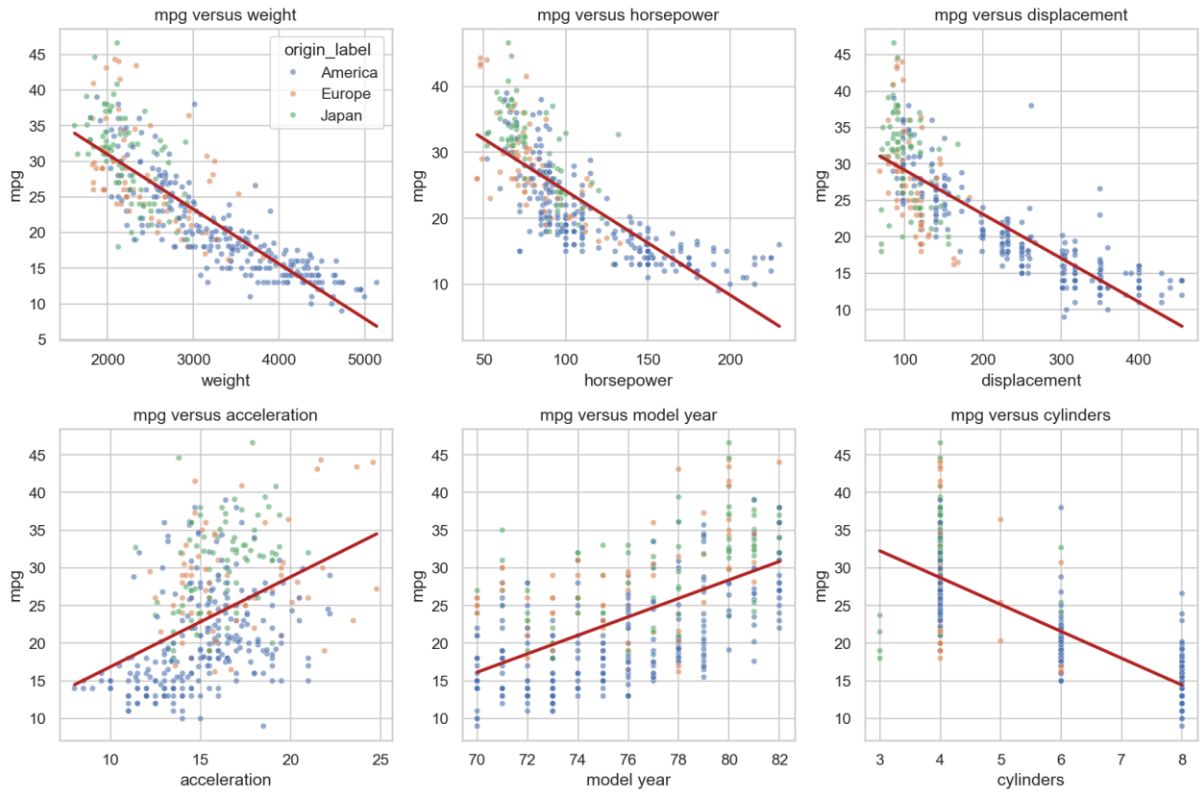


Figure 3: mpg against each input, coloured by origin, with a straight reference line to test curvature

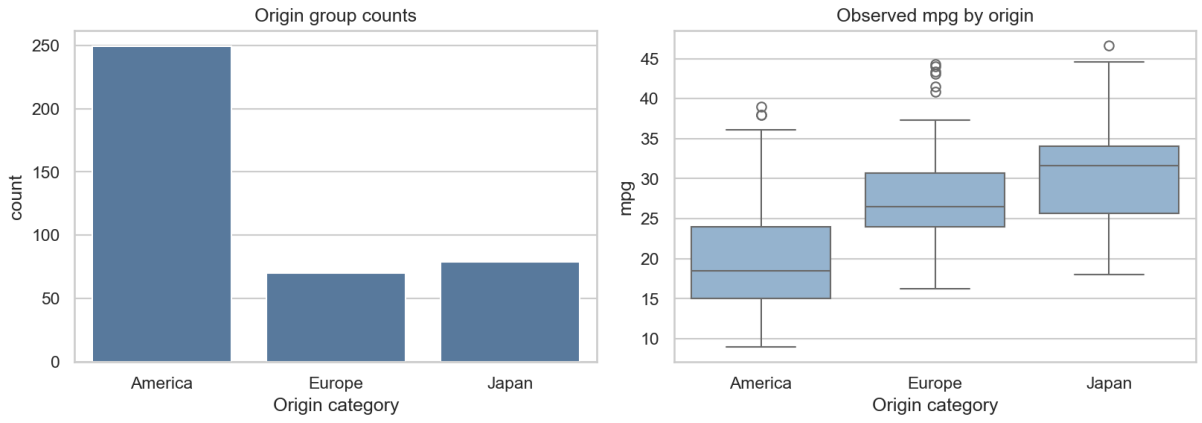


Figure 4: Counts and observed mpg by origin after restoring the labels

Implications for practice

A text-loaded column can hide missing values behind a token; parsing plus a range check is a professional test. Treating origin 1 < 2 < 3 as a scale places Europe midway between America and Japan; one-hot encoding removes that order. Legally, UCI's CC BY 4.0 release needs attributing (Quinlan, 1993), as the house data need De Cock (2011). Socially, these are US-market 1970 to 1982 cars; fuel-economy claims today would extrapolate and raw origin comparisons mislead.

Conclusion

Six '?' strings hid missingness from a default load, the main finding. EDA should parse types, check ranges, count pairs and keep nominal codes nominal before modelling.

References

De Cock, D. (2011) 'Ames, Iowa: alternative to the Boston Housing Data as an end of semester regression project', *Journal of Statistics Education*, 19(3). Available at: https://doi.org/10.1080/10691898.2011.11889627

Géron, A. (2025) *Hands-On Machine Learning with Scikit-Learn and PyTorch*. Santa Rosa: O'Reilly.

Quinlan, R. (1993) *Auto MPG* [Dataset]. UCI Machine Learning Repository. Available at: https://doi.org/10.24432/C5859H

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